# MILITARY SPECIFICATION

## WINCHES, DRUM, HAND-OPERATED SAFETY TYPE, FOR PONTOON BARGES

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers hand-operated winches for use on pontoon barges.

1.2 Classification. Winches shall be of the following types and sizes, as specified (see 6.2).

Type I - Anchor handling.
Size 5 - 10,000 pounds working load.

Type II - Deck machinery.
Size 2 - 4,000 pounds working load.

2. APPLICABLE DOCUMENTS

SPECIFICATION

FEDERAL

RR-W-410 - Wire Rope and Strand.
TT-P-664 - Primer Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116 - Preservation and Packaging, Methods of.
MIL-P-514 - Plates, Identification, Instruction and Marking Blank.
MIL-T-704 - Treatment and Painting of Material.
MIL-P-46093 - Primer Coating, Synthetic (for Brake Drums).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3950

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STANDARD

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A36 - Specification for Structural Steel.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

Technical society and technical association specifications and standards are generally available among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Description. Winches shall be complete with wire rope installed on the drums. Removable drive-nut crank or handwheel, rope end load hook or hook attachment provision, and 4-bolt drilling pattern for mounting, as specified herein.

3.2 First article. When specified (see 6.2), the contractor shall furnish a winch for first article inspection and approval (see 4.2.1 and 6.4).

* 3.3 Standard commercial product. The equipment shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the unit being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model(s).

* 3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercail practice. Unless



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otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

* 3.5 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable, and shall be designed to facilitate maintenance and repair operations.

* 3.6 Cleaning, treatment, and painting. Unless otherwise specified (see 6.2), surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. When specified (see 6.2), the unit shall be cleaned, treated, and painted in accordance with MIL-T-704, type A. The color of the finish coat shall be as specified (see 6.2). Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants, such as soil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.7 Identification marking. An identification plate and a transportation data plate conforming to MIL-P-514 shall be permanently affixed to the winch in a conspicuous place. Plate shall be of nonferrous type.

3.8 Instruction plates. The winch shall be equipped with instruction plates suitably located, describing any special or important procedure to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment, and shall be securely affixed thereto with nonferrous screws, bolts, or rivets of not less than 1/8-inch diameter.

3.9 Performance. The rated load shall be lifted smoothly, by application of not more than 10 pounds of continuous force to the crank handle in extended position of type II (2-ton capacity) winch, or 35 pounds of continuous force to the handle or handwheel of type I (5-ton capacity) winch. Except when lowering effort is applied to the crank or handwheel, 1.25 times the applicable working load shall not move downward. Application of the handbrake shall bring a falling load suspended by a type I winch to a smooth stop without grab or chatter. Type II winch shall have no noticeable backlash between gear teeth when reversing direction of the handcrank motion without load.

3.10 Design and construction. The winch shall be of steel construction, assembled as a unit. Unless otherwise specified (see 6.2), the base or adaptor frame shall be drilled with four 11/16-inch diameter holes. Bolt



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spacing shall be 16 inches center-to-center by 11-1/2 inches center-to-center, with the centerline of the winch drum mounted parallel to the long dimension. The base or base frame shall be flat for mounting to the steel deck or platform. Drive shaft ends shall be fitted with a crank or handwheel. Bearing surfaces shall be protected from entrance of water and wind-blown sand. Cranking handgrips shall rotate freely without binding, under normal operating conditions. Drum surfaces shall be smooth. Drum diameter shall be not less than 4 inches. Flange clearance shall be not less than 2 inches, with storage capacity of wire rope wound on, and anchored to the drum. Wire rope shall conform to RR-W-410, type I, class 2, 6 by 19 extra, improved plow steel, with characteristics in accordance with table XI. Rope ends shall be seized.

3.10.1 Type I - anchor handling. The type I winch shall have a horizontally mounted drum with overwind wire rope payout, and shall be complete with not less than 160 feet of 5/8-inch nominal diameter wire rope and a 10,000 pound rated capacity hook attachment provision. The winch shall be provided with either a 2-position crank with adjustments in handle length between 12 and 24 inches, and not less than 2-speed gearing with speed change by replacement of the crank on intermediate gear drive shafts; or a circular handwheel of 24-inch diameter with cranking handgrip 12 inches from the axle, and not less than 2 speeds with shifting provisions. The gear train shall incorporate an automatic safety-type load lowering brake of the self-energizing type, released only by lowering effort applied to the drive shaft; and a band type handbrake, with locking provision. The automatic brake shall have a ratchet release provision for free spooling, with control by handbrake application. The automatic load brake shall be designed so that load tension tightens the clutch pressure plate, and downward effort applied to any gear train drive shaft results in controlled downward motion. The automatic brake may be installed in the crank handle in lieu of the gear train. Removal of brake installed handles for shifting shall not result in lost control of the load.

* 3.10.2 Type II - deck machinery. The type II winch shall have a horizontally mounted drum, with overwind wire rope payout, and shall be complete with not less than 200 feet of 1/2-inch nominal diameter wire rope and 4,000 pound rated capacity hook attachment provision. The drum shaft shall be keyed to a driven gear. The drive shall have 2-speed gearing with speed change by selection of drive shaft; and a removable one-piece crank, approximately 14 inches long.

3.11 Lubrication. Lubrication means shall be provided for all moving parts requiring lubrication. Pressure lubrication fittings shall not be employed where normal lubrication pressure may damage grease seals or other parts, unless pressure relief is provided. The winch shall be lubricated prior to delivery, and shall be tagged in a conspicuous place to identify the lubricants used and their operating temperature ranges.

* 3.12 Workmanship.

3.12.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

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3.12.2 Bolted connections. All bolts shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided. All nuts, bolts, and screws shall be tight.

3.12.3 Riveted connections. All rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall completely fill the hole. Rivet heads shall be of an approved shape and shall be concentric with the rivet. The rivets shall be so fabricated as to develop a joint strength not less than the design value.

3.12.4 Welding. Surfaces to be welded shall be free from foreign matter which would be injurious to the weld. Welding procedures shall be in accordance with a nationally recognized code. Welds shall be of sufficient size and shape to develop the full design strength of the parts connected by the welds. Welds shall transmit imposed stresses without permanent deformation or failure when subjected to proof or service loadings.

3.12.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.2.1).
(b) Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. First article inspection shall be performed on one winch when a first article sample is required (see 3.2). This inspection shall include the examination of 4.3 and the test of 4.4. The first article may be standard production item from the supplier's current inventory provided the winch meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining winches to be furnished under the contract.

4.2.2 Quality conformance inspection. This inspection shall include the examination of 4.3, the applicable test of 4.4, and the requirements of 4.5.

* 4.3 Examination. Each winch shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the supplier's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements, shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual



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examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 Tests. Winches shall be tested in accordance with 4.4.1 and 4.4.2 as applicable to the type. All defects shall be corrected. Uncorrectable defects shall be cause for rejection.

4.4.1 First article test. First article winches, when included in the contract (see 3.2), shall be tested for all applicable performance characteristics by use of wire rope rigging arranged to provide rated working load tension, and 1.25 times rated working load tension, as measured between the winch and the first fairlead.

4.4.2 Production winch tests. Production winches and working parts shall be assembly line tested for fit and function. Source and characteristic data as proof of conformance, for materials and items not manufactured by the supplier and for conformance to referenced documents, shall be available to the contracting officer.

4.5 Packaging inspection. The preservation, packaging, packing, and marking of the winches shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1 Preservation - packaging. The preservation - packaging shall be level A or commercial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Disassembly. Disassembly shall be the minimum necessary to safeguard parts known to be subject to damage or loss and to accomplish reduction in cube.

5.1.1.2 Preservatives. Preservatives shall conform to the applicable specifications listed in, and shall be applied in accordance with MIL-P-116.

5.1.1.3 Unpainted surfaces. Unpainted exterior ferrous metal surfaces of component parts, such as nuts, bolts, ratchet, pawl, and any other unpainted metal surfaces, shall be coated with type P-1 preservative. Wire rope and fittings shall be coated with preservative as specified herein, wound on the drum, and secured to prevent unwinding.

5.1.1.4 Brake. The brake drum facing shall be coated with a thin film of primer conforming to TT-P-664 or MIL-P-46093.

5.1.1.5 Exposed gears. Exposed gears shall be coated with type P-1 preservative, or a thin film of primer conforming to TT-P-664 or MIL-P-46093.

5.1.1.6 Repair parts and maintenance tools. Repair parts shall be preserved and packaged in accordance with MIL-P-116. Unprotected metal surfaces of maintenance tools shall be coated with type P-1 preservative.

5.1.1.7 Technical publications. Technical publications for each winch shall be preserved in accordance with MIL-P-116, method IC-3.



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5.1.1.8 Consolidated packaging. Repair parts, tools, and technical publications shall be packaged in close fitting boxes conforming to PPP-B-636, class weather-resistant.

5.1.2 Commercial. The complete winch shall be preserved and packaged in accordance with the supplier's standard practice.

5.2 Packing. The packing shall be level A, B, or commercial, as specified (see 6.2).

5.2.1 Level A. Each complete winch shall be packed in a box conforming to PPP-B-621, class 2; or PPP-B-601, overseas type. The contents shall be blocked and braced to prevent movement inside the container or damaging of contents.

5.2.2 Level B. Each complete winch shall be packed as specified for level A, except that boxes shall conform to PPP-B-621, class 1, or PPP-B-601, domestic type.

5.2.3 Commercial. The complete winch shall be packed in a manner which will insure arrival at destination in satisfactory condition. Packing shall comply with applicable carrier rules and regulations.

5.3 Marking. Interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The winches specified herein are primarily used on pontoon barge assemblies for anchor and miscellaneous cable control, and are usable anywhere that a bolted base mounting can be prepared.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of winch required (see 1.2).
(c) When a first article is required for inspection and approval (see 3.2, 4.3.1, and 6.4).
(d) When cleaning, treatment and painting shall conform to MIL-T-704, type A (see 3.6).
(e) Color of finish coat required (see 3.6).
(f) Mounting bolt hole template other than as specified (3.10).
(g) Level of preservation and packaging and level of packing required (see 5.1 and 5.2).

6.3 Contract data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of paragraph 7-104.9(n) of the Defense Acquisition Regulations (DAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.



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6.4 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of paragraph 7-104.55 of the DAR. The first article should be a first production item consisting of one complete winch, or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The contracting officer should include specific instructions in all procurement instruments, regarding arrangement for examination, tests, and approval of the first article.

6.5 Revision changes. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-W-28547B | WINCHES, DRUM, HAND-OPERATED SAFETY TYPE, FOR PONTOON BARGES |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR<br>☐ USER<br>☐ MANUFACTURER<br>☐ OTHER *(Specify)*: |
| b. ADDRESS *(Street, City, State, ZIP Code)* | |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

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